

Worked Example — backend-swe-opt-triage.md

Scenario

Student: F-1, MS Information Systems, Northeastern University.
OPT start estimated August 2025. Backend SWE target (SOC 15-1252).
1 year Python/Node.js internship experience. Open to all company stages.

Input: 12 companies from LinkedIn job alerts ("Backend Engineer", "Software Engineer Backend", "Backend Developer") and a class referral list. No prior filtering applied.

Companies: Stripe, Plaid, Anysphere (Cursor), Temporal Technologies, Wander, Ramp, Rippling, Duolingo, Scale AI, ExampleGhostCo, Brex, Vercel

Mode Execution

Step 1 — Check pipeline and tracker

```
cat data/ats/pipeline.md
cat data/ats/applications.md
```

Result: None of the 12 companies appear in the existing pipeline or application log.
All 12 proceed to Step 2.

Step 2 — H-1B Petition Filter (SOC 15-1252 + 15-1244 + 15-1299, since 2022)

```
python SCRIPTS/h1b_filter.py --soc 15-1252,15-1244,15-1299 --min-petitions 2
```

Company	H-1B LCA Petitions (SWE SOC codes, 2022–2024)	Notes
Stripe	52	Consistent filings 2022–2024
Duolingo	31	Public; consistent
Rippling	24	Strong SWE history
Ramp	18	Growing year over year

Company	H-1B LCA Petitions (SWE SOC codes, 2022–2024)	Notes
Scale AI	14	Active filings
Brex	11	Down from 34 in 2019–2021; declining trend
Plaid	9	Lower than 2020–2021 peak
Vercel	3	Present but low volume
Temporal Technologies	2	1 under 15-1252; 1 additional under 15-1244
Anysphere (Cursor)	0	Founded 2022; no LCA history yet
Wander	0	No filings found
ExampleGhostCo	0	No filings found

Note: Running across three SOC codes surfaced one additional Temporal filing under 15-1244 that would have been missed by a single-code filter on 15-1252 alone.

Step 3 — Form D Recency Check (pre-IPO companies only)

```
python SCRIPTS/form_d_filter.py --company <name> --max-age-days 548
```

Company	Most Recent Form D	Amount	Flag
Anysphere (Cursor)	March 2024	\$60M	None — recent
Temporal Technologies	November 2023	\$75M	None — recent
Vercel	May 2023	\$150M	None — recent
Wander	April 2023	\$16M	FLAG — 26 months ago, small raise
ExampleGhostCo	January 2022	\$3M	FLAG — 41 months ago, very small raise

Step 4 — Job Liveness Check

```
node SCRIPTS/ats/liveness_check.js --url <job_url>
```

Company	Result
Stripe, Ramp, Rippling, Duolingo, Scale AI, Brex, Plaid	LIVE
Anysphere (Cursor), Temporal Technologies, Vercel	LIVE
Wander	EXPIRED
ExampleGhostCo	EXPIRED

Wander and ExampleGhostCo removed from consideration. 10 companies remain.

Step 5 — ATS Detection

```
python SCRIPTS/ats/ats_detect.py --company <name>
```

Company	ATS Detected
Stripe, Rippling, Duolingo, Brex	Greenhouse
Scale AI, Plaid	Lever
Ramp, Anysphere (Cursor), Temporal Technologies, Vercel	Ashby

All 10 remaining companies returned a known ATS provider. No unknown or stale career pages.

Step 6 — Cognitive Pivot Score

```
python SCRIPTS/cognitive_score.py --soc 15-1252
```

Result: SOC 15-1252 — Cognitive demand score 72/100. AI substitution risk: Moderate-Low. Score applies to the SOC code as a whole. Within-SOC differentiation (distributed systems vs. CRUD roles) requires manual role description review.

Step 7 — Final Triage Table

Company	H-1B SWE Petitions	Form D	ATS	Live?	Cog Score	OPT Tier
Stripe	52	N/A (public)	Greenhouse	LIVE	72/100	Tier 1
Duolingo	31	N/A (public)	Greenhouse	LIVE	72/100	Tier 1
Rippling	24	N/A (late-stage)	Greenhouse	LIVE	72/100	Tier 1
Ramp	18	N/A (late-stage)	Ashby	LIVE	72/100	Tier 1
Scale AI	14	N/A (late-stage)	Lever	LIVE	72/100	Tier 1
Brex	11	N/A (late-stage)	Greenhouse	LIVE	72/100	Tier 1*
Plaid	9	N/A (Series D)	Lever	LIVE	72/100	Tier 1*
Anysphere (Cursor)	0 (too new)	March 2024 – \$60M	Ashby	LIVE	72/100	Tier 2*
Vercel	3	May 2023 – \$150M	Ashby	LIVE	72/100	Tier 2
Temporal Technologies	2	Nov 2023 – \$75M	Ashby	LIVE	72/100	Tier 2
Wander	0	April 2023 – \$16M	Unknown	EXPIRED	72/100	Tier 3
ExampleGhostCo	0	Jan 2022 – \$3M	Unknown	EXPIRED	72/100	Tier 3

Tier 1* = Tier 1 with declining petition trend — requires manual verification before treating as full Tier 1.

Verified vs. Inferred

VERIFIED from scripts and data:

- H-1B petition counts (LCA disclosure data, multi-SOC filter)
- Form D recency and amount (SEC filings)

- Job posting liveness (Playwright via liveness_check.js)
- ATS provider (career page detection via ats_detect.py)

INFERRED — not verified from data:

- Whether Brex's declining petition count reflects a policy change or reduced hiring volume. The data shows a drop from 34 (2019–2021) to 11 (2022–2024). Could be layoffs. Could be intentional shift away from sponsorship. Requires manual Blind check before treating as Tier 1.
 - Whether Anysphere (Cursor) intends to sponsor. Zero LCA history because the company is only 2 years old — not because they have refused to sponsor. Tier 2* reflects uncertainty, not a negative signal.
 - Whether the role descriptions at Temporal and Vercel map to SOC 15-1252 or an adjacent code. The multi-SOC filter reduces but does not eliminate this ambiguity.
 - Within-SOC role quality for all 10 companies (cognitive score is SOC-level only).
 - OPT timeline compatibility for any company (proposed script does not exist yet).
-

What Was Missed on First Run

Brex petition decline not flagged automatically. The mode classified Brex as Tier 1 because it meets the ≥ 2 petitions threshold. A follow-up manual check (Blind threads, Layoffs.fyi) showed Brex conducted significant layoffs in 2022 and reduced international hiring for some teams. The data captures the decline in numbers but does not explain it. The correct action: apply to Brex but add a manual recruiter verification step before completing the full application. The mode was updated to add a Tier 1* flag for companies showing a year-over-year petition decline.

Single-SOC filter undercounted Temporal Technologies. First run used only SOC 15-1252 and returned 1 petition for Temporal. Running the multi-SOC filter (15-1252, 15-1244, 15-1299) returned 2, moving Temporal closer to the threshold and changing its Tier 2 classification from "marginal" to "reasonable." This is why the mode recipe now specifies the multi-SOC filter as the default for Step 2.

What I Would Change Based on Seeing the Output

Change 1 — Default to multi-SOC filter in Step 2.

The single-SOC run undercounted Temporal. The mode recipe now specifies:

```
python SCRIPTS/h1b_filter.py --soc 15-1252,15-1244,15-1299 --min-petitions 2
```

****Change 2 — Add Tier 1* for declining petition trend.****

The triage table now distinguishes Tier 1 (stable or growing petition count) from Tier 1* (verified history but declining year-over-year). Brex and Plaid both land in Tier 1* and require a manual verification step before applying. This distinction is currently judgment-based until the proposed `h1b_recency.py` script exists.

****Change 3 — Enforce manual verification as a logged step for Tier 2* companies.****

Anysphere (Cursor) is Tier 2* and currently the mode only flags it. A student in a time-pressured search may skip the manual verification. The log template now includes a required checkbox for Tier 2* companies before they move into the active pipeline:

```
- [ ] Manual sponsorship verification complete for all Tier 2* companies
```